

! Important: When the ‘Day-of-month’ and ‘Day-of-week’ fields are both other than *, the command will be executed when either of these fields match. This can be used, for example, to run a command on the fifteenth of every month, and on every Friday.

The first five of these fields all use the same syntax rules: * for “Don’t Care”/ always.

There is a number to specify the number of minutes or hours, a date, or a weekday. (For weekdays, 0 equals Sunday, 1 equals Monday, 2 equals Tuesday, etc. 7 also equals Sunday. x-y is used for a range, inclusive of x and y.) x, y for lists: Lists can include ranges as well, e.g., 5, 10-13, 17 in the ‘Minutes’ column, to indicate that a job should run at 5 minutes past the hour, 10 minutes past, 11 minutes past, 12 minutes past, 13 minutes past, and 17 minutes past.

*x indicates an interval of x, e.g., */7 in the ‘Minutes’ column will run a job in exactly every seven minutes.

Additionally, three-letter English abbreviations can be used for both months and weekdays, e.g., *Jan*, *Feb* and *Tue*, *Wed*.

The last field contains the command to be executed.

This command will be executed by */bin/sh* unless a SHELL environment variable has been declared. If the command contains an unescaped percentage sign (%), that sign will be treated as a new line, and everything after the percentage sign will be fed to the command or *stdin*.

Examples of cron jobs:

```
Start the crontab editor:
# crontab -e
Insert the following line:
* * * * * * date >> /home/student/my_first_
cron_job.
Save your changes and quit the editor.
```

```
Inspect all of your scheduled cron jobs.
# crontab -l
Wait for your job to run, then inspect the contents of the /
home/student/my_first_cron_job.
# cat ~/my_first_cron_job.
Remove all of the cron jobs for student.
# crontab -r.
```

Scheduling system cron jobs

System cron jobs: Apart from user cron jobs, there are also system cron jobs. System cron jobs are not defined using the *crontab* command, but are instead configured in a set of configuration files. The main difference in these configuration files is an extra field, located between the *Day-of-week* field and the *Command* field, specifying under which user a job should be run.

System cron jobs are defined in two locations: */etc/crontab* and */etc/cron.d/**. Packages that install cron jobs should do so by placing a file in */etc/cron.d/*, but administrators can also use this location to more easily group related jobs into a single file, or to push jobs using a configuration management system.

There are also predefined jobs that run every hour, day, week and month. These jobs will execute all scripts placed in */etc/cron.hourly/*, */etc/cron.daily/*, */etc/cron.weekly/*, and */etc/cron.monthly/*, respectively. Do note that these directories contain executable scripts, and not cron configuration files.

! Important: Make sure that any scripts you place in these directories are made executable. If not (e.g., with *chmod+x*), the task will not run as per the schedule.

The */etc/cron.hourly/** scripts are executed using the *run-parts* command, from a job defined in */etc/cron.d/0hourly*. The daily, weekly and monthly jobs are also executed using the *run-parts* command, but from a different configuration file: */etc/anaacrontab*.

In the past, */etc/anaacrontab* was handled by a separate daemon (*anacron*), but in Red Hat Enterprise Linux 7, the file is parsed by the regular *crond* daemon. The purpose of this file is to make sure that important jobs will always be run, and not skipped accidentally because the system was turned off or hibernating when the job ought to have been executed.

The syntax of */etc/anaacrontab* is different from the other cron configuration files. It contains exactly four fields per line:

```
Period in days
Once per how many days this job should be run
Delay in minutes
The amount of time the cron daemon should wait before
starting this job
Job Identifier
```

This is the name of the file in */var/spool/anacron/* that will be used to check if this job has run. When *crond* starts a job from */etc/anaacrontab*, it will update the timestamp on this file. The same timestamp is used to check when a job has last run in the past.

Command: This is the command to be executed. */etc/anaacrontab* also contains environment variable declarations using the syntax *NAME=value*. Of special interest is *START_HOURS_RANGE*. Jobs will not be started outside of this range.

Managing temporary files with system-tmpfiles

A modern system requires a large number of temporary files and directories. Not just the highly visible ones such as */tmp* that get used and abused by regular users but also

Continued on page no...40